

The Mystery of Agency*

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Abstract. I discuss four puzzles about how we tend to think of an agent’s choices, when we evaluate them from the agent’s own perspective. The puzzles suggest that agency involves a luminous and infallible capacity to bring about special “willings”. It is hard to believe that real people have such a mysterious capacity. I argue that this is nonetheless a central ingredient in our cognitive model of how people make choices. When we ask whether someone has an ordinary ability, we therefore take for granted that they can bring about the relevant “willings”.

1 Introduction

We humans think a lot about agency – about what people do, about what they can do, and what they ought to do. I want to highlight four puzzles raised by the way we tend to approach these questions. None of the puzzles is new, but they are usually discussed in isolation; I will argue that they have a common source and a common solution.

The first puzzle, to be discussed in sections 2 and 3, arises from two features of the “perspectival ‘ought’”. On the one hand, the perspectival ‘ought’ appears to supervene on the agent’s perspective or evidence. On the other hand, this sense of ‘ought’ seems to imply ‘can’. But couldn’t an agent lack information about what they can do?

My second puzzle (section 4) concerns the description of the available acts. If we want to evaluate an agent’s choice from their perspective, the options must be described in a way that does not go beyond the agent’s evidence. But what ensures that the agent has sufficient information to identify and distinguish their options?

My third puzzle (section 5) is that we tend to evaluate an agent’s options as if they were uncaused interventions. We consider the consequences of the options but not their causes. But how could someone bring about an uncaused intervention?

This puzzle has a famous sibling. Normal people can’t influence the laws of nature or the distant past. It seems to follow that they only have a choice if their actions are undetermined. But we don’t view rational choices as random chance events either. How can an agent’s choices be neither determined nor random? This is my fourth puzzle.

* Thanks to Mahrad Almotahari, Beri Marušić, Milo Phillips-Brown, Tom Schoonen, Patrick Todd, Barbara Vetter, and the participants of the *Epistemology of Ability* workshop at HU Berlin in 2024 for helpful comments on earlier versions.

Together, the four puzzles will lead us to a certain picture, a picture in which agents have an infallible and luminous capacity to “will” various things, unconstrained by the circumstances and the natural order. Explicitly stated, the picture seems crazy. I’ll argue (in sections 7 and 8) that it is, nonetheless, *almost* correct. In section 9, I will explain how all this might relate to the epistemology of abilities.

2 ‘Ought’ and ‘can’

Let’s begin with the first puzzle. We often evaluate choice situations from the agent’s perspective, bracketing external facts of which the agent may be unaware. I believe, for example, that people should sometimes use a map to find their way, even though it would be better “objectively”, in light of all the facts, to take the correct turns without wasting time on consulting the map. Similarly, I believe that physicians should examine their patients, even though it would be better objectively to skip the examination and immediately start with the optimal treatment. Physicians, gamblers, military commanders, and stock traders have sophisticated rules for how to act given limited information.

When we assess a choice situation in this fashion – wondering what the agent ought¹ to do given their limited information – we engage in a *perspectival evaluation*, as I’ll call it. The term is unfamiliar, but the activity is not. We do it all the time.

A defining feature of perspectival evaluation, it seems, is that it does not draw on facts beyond the agent’s perspective: if two possible agents in two decision situations have the same perspective, a perspectival evaluation of their situation would yield the same verdict. This *perspectival supervenience* is the first ingredient in my first puzzle.

The second ingredient is that the perspectival ‘ought’ seems to imply ‘can’. Suppose Jones is lost in the woods. We wonder what he should do. You suggest that he should use the map on his phone to find a way out. I point out that his phone is out of battery. It would be absurd to insist that he should nonetheless consult the map on his phone. (Perhaps he should *try* to use the map. This he can do.) If we know that an option isn’t available, it is ruled out as an answer to our perspectival evaluation of what the agent ought to do.

Let’s get clear on the notion of ‘can’ that is at issue. We may assume that Jones has the “general ability” to navigate with his phone. He *can* use the map on his phone, in the sense in which a pianist can play the piano even if no piano is around. But there’s another sense in which he can’t. If we judge that an agent in a particular choice situation ought to do so-and-so, we assume that it is entirely up to them whether they do so-and-so: all internal and external preconditions for the act are in place. Let’s call this kind of ‘can’ – the ‘can’ that seems to be implied by the perspectival ‘ought’ – the *practical ‘can’*.

Now here’s the puzzle. If an agent’s perspective settles what they ought to do, and ‘ought’ implies ‘can’, then their perspective must settle that they can perform the relevant act. But couldn’t an agent lack information about what they can do (in the practical sense of ‘can’)?

¹ I’ll keep using ‘ought’, but the word is not important. Some philosophers have reasoned themselves into a position where they can no longer see that a physician ought to examine their patients. I hope you can at least agree that examining the patients is in some sense the right (or rational) thing to do.

I here assume that an agent's perspective is related to their information. Let's make this explicit.

Intuitively, an agent's perspective comprises what is "given" or "accessible" to the agent. Perspectival evaluation does not draw on "external" facts that are not given. These are metaphors; the puzzle arises in almost any way of unpacking them. For concreteness, I will assume the following connection between perspective and evidence/information about practical abilities:

Link. If an agent's perspective settles that they have (or lack) some practical ability then their evidence entails that they have (or lack) the ability.

Suppose, for example, that Jones's evidence leaves open whether he can climb the tree in front of him. By *Link*, it follows that his perspective does not settle that he can climb the tree: there is a possible agent with the same perspective who is not able to climb the tree.

The puzzle now arises as follows. Assume an agent ought to ϕ , in the perspectival sense of 'ought'. By perspectival supervenience, the agent's perspective is enough to settle that they ought to ϕ . By 'ought'-implies-'can' (and the fact that anything that is entailed by something that is settled by the agent's perspective is also settled by the agent's perspective), the agent's perspective settles that they *can* ϕ . By *Link*, the agent's evidence entails that they can ϕ .

The puzzle can be strengthened in several ways.

First, 'ought' doesn't just imply 'can', it plausibly also implies 'can't do better'. If, in a certain chess game, it would be better to move the rook rather than the bishop, then it could hardly be true that the player ought to move the bishop. If the perspectival 'ought' supervenes on the agent's perspective, it follows that the agent's perspective – and thereby their evidence – must settle not only that they *can* do whatever they ought to do, but also that they *can't* do other things that would be better if they were available.

Second, perspectival evaluation allows for judgements not just about what an agent *ought* to do, but also about what they *may* do, what would be *supererogatory*, or what they *should do given that* they are determined not to do so-and-so. In each case, the answer seems to imply that the agent has the practical ability to perform the relevant act. So the agent's evidence must entail that they have this ability.

Third, our practice of perspectival evaluation has a slot in which we can plug in norms or goals. We can ask what an agent ought to do *by the lights of hedonistic utilitarianism*, or *by Kantian standards*, or *in order to promote their personal wealth*.² In each case, saying that the agent ought to ϕ seems to imply that they have the practical ability to ϕ . Since different norms or goals often select different acts, the agent's evidence must settle that all these acts are available.

In sum, our practice of perspectival evaluation seems to presuppose that agents have detailed information about their practical abilities. This is puzzling. Couldn't an agent lack this information? Don't most people, in fact, have imperfect information about what they can do?

If an argument leads to a patently false conclusion, we need to reconsider its premises. We might deny that the perspectival 'ought' (and 'may', etc.) supervenes on the agent's perspective. But this seems to go against the very idea of perspectival evaluation. Perhaps, then, we should deny that the

² Some reserve the term 'perspectival' for evaluations that are tied to the agent's personal goals. The label is not important.

perspectival ‘ought’ (etc.) implies ‘can’?³ The implication would have to fail across the board, not just in esoteric cases. This also looks unattractive. Before we jump to either conclusion, let’s take a closer look at a famous example.

3 Willings

Remember Jill the physician, from [Jackson 1991]. Jill’s patient suffers from a minor skin condition. She has a choice between three drugs. Drug A would relieve the symptoms but not cure the condition. Of drugs B and C, one would cure the condition (without side effects), the other would kill the patient. Jill doesn’t know which of B and C would have which effect, and she can’t acquire the information. Let’s say that B is the cure and C is the killer. Objectively, it would be best to give drug B. Given Jill’s information, however, the right choice is drug A.

Now suppose that, unbeknownst to Jill, someone has replaced the drug in the packaging of drug A with drug C, discarding the real drug A. As a result, Jill has false beliefs about what she can do. She believes that she can give drug A, but she can’t. What ought she to do? Is it still true that she ought give drug A?

I wouldn’t say so. What she ought to, I’d say, is something like *giving the drug in the packaging of drug A*. Jill might describe this act as ‘giving drug A’, but her description would be false.

Information about the unavailability of drug A influences my judgements in a way that information about, say, unknown side effects of drug A do not. In Jackson’s original scenario, where giving drug A is an option, I’m happy to say that Jill ought to give drug A even if, unbeknownst to her, the drug has side effects that make it better to give no treatment at all. If drug A is not available, by contrast, I want to redescribe the option. (I hope you do, too.) I want to describe it in a way that preserves the link between ‘ought’ and ‘can’. Jill can give the drug in the packaging of drug A, and this is what she ought to do.

Does this mean that we should give up perspectival supervenience? The perspectival ‘ought’ appears to be sensitive to whether someone has, *unbeknownst to Jill*, tampered with the drugs. In Jackson’s original scenario, Jill ought to give drug A. In my modified scenario, she instead ought to give the drug in the packaging of drug A. Yet her perspective remains the same.

One might, however, put pressure on the judgement about the original scenario. Focus on that scenario. Nobody has tampered with the drugs. But presumably this isn’t entailed by Jill’s perspective. – Otherwise there would be no threat to perspectival supervenience. Jill’s perspective is compatible with my modified scenario. It is also compatible with, say, a scenario where drugs A and C have been swapped. These possibilities are *unlikely* in light of Jill’s perspective or evidence, but they are not impossible. Let’s say that there’s a 0.01% probability that drugs A and C have been swapped, given Jill’s evidence. As a consequence, Jill’s evidence doesn’t settle what she would have to do in order to give drug A: which packaging she would have to open. Now reconsider the question what she ought to do. Ought she to give drug A? I’m reluctant to say so. In our practice of perspectival evaluation, it seems wrong to judge that an agent ought to do something if they lack information about how they could do it. I’d rather say that Jill ought to *give the drug in the packaging of drug A*.

3 This is how [Graham 2011: p.340] and [Way and Whiting 2016: p.1888] respond to the present puzzle.

A third answer to our puzzle now comes into view. We can accommodate ignorance of one's practical abilities without denying either perspectival supervenience or the connection between normative judgements ('ought') and practical ability ('can'). If an agent doesn't know whether they can do so-and-so, *doing so-and-so* is not a suitable object of perspectival evaluation: it isn't a candidate for something the agent ought to do. On closer inspection, the premises of our puzzle do not entail that agents must have comprehensive information about their practical abilities. It is enough that anything an agent can do is identifiable by the agent under a description (like, perhaps, *giving the drug in the packaging of drug A*) which they know they can make true.

But where does this lead us? In a real-life scenario, Jill's perspective arguably wouldn't settle that she can give the drug in the packaging of drug A. Couldn't the package be empty? Couldn't the security forces, or the laws of nature, prevent her from opening the package? Couldn't she be about to have a stroke or seizure, losing control of her arms? These possibilities may be far-fetched, but arguably they aren't strictly ruled out by her perspective: any of them might obtain while her perspective remains the same.

Take the seizure possibility. Many people have undiagnosed epilepsy. Even if Jill has never experienced symptoms, it would be odd to think that her evidence conclusively rules out this possibility. For all she can tell, she might be about to have a seizure that would prevent her from opening any packaging. If she *will* have a seizure, it's not true that she ought to open some packaging. So her evidence doesn't settle that she ought to do it. Indeed, when I attend to the seizure possibility, I'm inclined to say that Jill ought to *try to* open the packaging of drug A, rather than that she ought to open the packaging.

We are led to the conclusion that the options that figure in perspectival evaluation are typically not overt acts like *giving drug A* or *giving the drug in the packaging of drug A*, but special acts of *trying, intending, deciding*, or *willing* whose practical availability is not in doubt.⁴

You may wonder how agents are supposed to know which of these special acts they can perform. Can't we lack information about what we can try or will? I'll come back to this worry. You may also object that our practice does not align with the conclusion I've reached. We rarely talk about what agents should try or will. We say that a chess player should *move their rook*, or that a physician should *examine their patient*; Jackson says that Jill ought to *give drug A*. Let me briefly address this second objection.

When we evaluate an agent's choice, we normally take for granted certain facts about the connection between their narrowly construed options – the available “willings”, as I'll say from now on – and overt acts, even if these facts are not strictly entailed by the agent's perspective or evidence. We ignore the possibility that the agent might have an epileptic seizure. If such possibilities are set aside, the agent's evidence does entail that they can perform ordinary, overt acts. When we think about Jackson's scenario, we also set aside the possibility that someone has replaced the drugs in their packaging. Jill's evidence entails that she gives drug A in every *non-ignored* world in which she wills to give the drug in the packaging of drug A.

Now suppose the perspectival 'ought' and 'can' are context-dependent in roughly the way [Lewis

⁴ This idea has been defended, on more or less the present grounds, in [Ross 1939], [Prichard 1949], [Zimmerman 2008: ch.3], and [Hedden 2012].

1996] argued that ‘know’ is context-dependent. The specifics depend on independent questions about how these expressions work, but the basic idea is to restrict all quantification over worlds to worlds that aren’t “properly ignored” in the conversational context.⁵ ‘Jill ought to give drug A’ is true in Jackson’s original scenario – but only if we ignore, for example, the far-fetched possibility that someone might have swapped the drugs. It becomes false if attention is drawn to this possibility. It is also false in my modified scenario, where the far-fetched possibility is actual: the actual world is never properly ignored.

There’s more to say about this, but I have other puzzles to get to. Let me wrap up. We began with an apparent tension between perspectival supervenience and ‘ought’-implies-‘can’. Instead of dropping one of these assumptions, I have argued that our practice of perspectival evaluation assumes a special domain of “willings” as ultimate objects of choice. There remains a puzzle. What are these willings? Why would agents always have perfect information about which of them they can perform?

4 Expectations

I have said little about how the perspectival evaluation of an agent’s options works. How do we assess what an agent should do, given their evidence or perspective? A familiar answer looks at the possible outcomes (broadly understood) of each option and adds up their value, perhaps weighted by their probability (given the agent’s evidence). Whatever the details, this approach only yields sensible results if the agent’s options are individuated in a certain way.

Return to Jackson’s scenario. Jill knows that one of drugs B and C would cure her patient and the other would kill him. Since drug B is the actual cure, an act of *giving drug B* can also be described as an act of *curing the patient*. But this is not an adequate description for assessing Jill’s options. *Curing the patient* is certain to have the best outcome. If it was one of Jill’s options, we could hardly say – as we should – that she ought to give (or will to give) drug A.

An adequate description of an agent’s options must not draw on facts outside the agent’s evidence. The options must be described in a way that is *transparent* to the agent, we might say. What exactly does this require?

One might suggest that the options must be described in a way of which the agent’s evidence settles that they can perform the so-described act. But this is not enough. Ignoring far-fetched possibilities, Jill’s evidence settles that one of her options amounts to curing the patient. She knows that she can cure him. But she doesn’t know how.

An adequate option description must leave no room for such ignorance. If ϕ is an adequate option description, the agent must have no rational doubt about which choice would amount to ϕ ing. As a corollary, the agent should be in a position to become rationally certain that they are ϕ ing merely by making a decision.

Clearly, overt acts usually don’t meet this condition. In a real situation, Jill could hardly be certain

⁵ For instance, a conditional analysis of the practical ‘can’ might say that ‘*S* can ϕ ’ is true iff *S* ϕ s in the “closest” non-ignored worlds in which *S* wills to ϕ . Similarly, if we start with the hypothesis that ‘*S* ought to ϕ ’ is true (on its perspectival reading) iff *S*’s evidence entails that the closest worlds in which *S* ϕ s are at least as good (by the relevant standards) as the closest worlds in which *S* performs any other act that *S* can perform, then on the revised approach, we would restrict both the evidence and the relevant closest worlds to ones that aren’t properly ignored.

that she will be giving drug A merely on the basis of making a decision. As we noted before, her evidence doesn't settle that the packaging contains the drug, or that her arms will keep working. A full evaluation of her decision problem should take these possibilities into account. If there is a small but positive probability, given Jill's evidence, that drug A has been replaced with a lethal alternative, then *giving drug A* is not an adequate description of the option, as it would rule out the possible outcome of killing the patient.

For ordinary acts of ordinary agents, there is always an epistemic gap between decision and act: the agent's evidence doesn't settle that the act will take place if the decision is made. The transparency requirement therefore suggests that the options shouldn't be described as ordinary acts, but as special acts of "trying" or "willing" for which no such gap can arise: when you decide to will, you can be certain that the willing takes place.⁶

The conclusion we've reached matches the conclusion from the first puzzle. But the argument is different. The earlier argument relied on perspectival supervenience and 'ought'-implies-'can'. The present argument only assumes the transparency of options.

As before, a puzzle remains. What are these special acts of trying or willing? Why should they meet the transparency condition? Isn't there always a gap between deciding to ϕ and ϕ ing, no matter what ϕ might be?

5 Interventions

My next puzzle turns on another aspect of how we evaluate an agent's options. Suppose an agent has a choice between some options. Let's assume these are ordinary acts – the puzzle would emerge just as well if they were willings. The puzzle is that in order to figure out what the agent ought to do, we only consider what might happen *as a result* of their choice. We reason forward from the hypothetical choice, not backward.

To see what I mean, consider poor Jack, who needs to go to town. He could take the bus for £2, as usual, or he could hire a limousine with a personal driver for £2,000. He's not going to hire the limousine, of course. He would do so only if he were immensely rich, which he is not. Someone might suggest that Jack *should* take the limousine, on the grounds that – as I just said – he would be immensely rich if he did. This would be reasoning backward, from the hypothetical choice to earlier circumstances that might have brought it about. It's not how we evaluate options.

When we evaluate what would happen if Jack were to take the limousine, we don't ask why he would have made this choice. We don't assume that the relevant limousine worlds are worlds where he had good reason to take the limousine. Nor do we assume that they are worlds in which Jack has lost his mind, due to a brain tumor, say: We don't think he should take the bus because otherwise he'd have a brain tumor. We seem to evaluate an agent's options as *interventions* that affect the world's causal structure "from the outside", without explicable causes.⁷ This is puzzling. For we know that choices have causes. They are not uncaused interventions.

⁶ That options should be construed as "acts of the will" is a popular view among decision theorists, essentially because of the transparency condition. See, for example [Sobel 1983], [Weirich 1983], [Joyce 1999: ch.2]. See also [Holguín and Lederman 2024].

⁷ The hypothesis that an agent's options should be evaluated as uncaused interventions, at least from the agent's own

We can sharpen the puzzle. Suppose Jocasta believes that the universe conforms to certain deterministic laws L .⁸ Today, she is asked if it does, and answers ‘yes’. Assuming her only relevant aim is to tell the truth, this is intuitively the right choice. She is confident that answering ‘yes’ will achieve her aim. But what would have happened if she had answered ‘no’? Or rather, what do her beliefs imply about what would have happened?

Let w be an arbitrary world of the kind Jocasta believes she inhabits: a world that conforms to L and where she answers ‘yes’. Since L is deterministic, any world in which she answers ‘no’ either doesn’t conform to L or differs from w in its entire history.⁹ By the criteria of [Lewis 1979], the “closest” worlds to w at which Jocasta answers ‘no’ are worlds that don’t conform to L . According to Lewis, Jocasta would have spoken the truth if she had said ‘no’. We don’t get the intuitive result that Jocasta was right to answer ‘yes’.¹⁰

[Dorr 2016] and others have argued that the relevant closeness measure should hold fixed the laws and vary the past: if Jocasta had answered ‘no’ the world would still have conformed to L but its entire history would have been different. But suppose Jocasta believes that some event E took place in the past, and she is asked whether it took place. She answers ‘yes’. If E involves an aspect of the past that would have been different if she had said ‘no’¹¹, it follows that Jocasta would have said something true, and we don’t get the intuitive result that answering ‘yes’ was the right choice.¹²

The problem is that neither the distant past nor the laws of nature are intuitively a *result* of Jocasta’s present choice. In a deterministic world where Jocasta answers ‘yes’, the laws and the past are pre-conditions that *lead* to her choice. Since our practice of perspectival evaluation requires reasoning forward from the hypothetical choice and not backward, we want to hold fixed the laws and the past when we evaluate Jocasta’s options. But this looks impossible. If we hold fixed the laws and the past, and the laws are deterministic, how can we vary Jocasta’s choice?

This puzzle is obviously related to a fourth one, canonized by [van Inwagen 1983]. The fourth puzzle starts with the plausible assumption that the distant past and the laws are outside our control: there is nothing ordinary people can do that could make any difference to the past or the laws. For any choice we can make today, if we were to make it, the distant past and the laws would be just as they actually are. If the laws are deterministic, it seems to follow that the only choices we *can* make are the ones we *will* make.

perspective, has a venerable history. See, for example, [Kant 1781: A542/B570–A557/B585], [Ramsey 1931], [Ismael 2007], [Solomon 2021].

8 I use ‘deterministic’ in the standard Montagovian sense in which a system of laws is deterministic iff any two worlds that perfectly obey the laws are “alike always or never” [Lewis 1979: 37]; compare [Montague 1974], [Earman 1986: 20f.]. This is stronger than Steward’s use in her chapter of the present volume.

9 This follows from the definition of determinism, see the previous footnote.

10 Worse, suppose Jocasta is not 100% certain that the world conforms to L . In worlds where L is false, her utterance of ‘yes’ does not achieve her aim of telling the truth, but an utterance of ‘no’ would almost certainly have done so. In worlds where L is true, both utterances achieve her aim. On balance (in expectation), we have to say that Jocasta should have said ‘no’! See [Ahmed 2013].

11 On Dorr’s view, it would be hard for Jocasta to have evidence (in worlds like ours) for an aspect of the past that depends on her present choice. But we can evaluate her choice in light of her beliefs, without assuming that her beliefs are based on evidence.

12 As above, we can easily get the opposite result that Jocasta should have answered ‘no’, if Jocasta isn’t 100% certain about E .

This puzzle can't be avoided with a compatibilist account of freedom. The puzzle is not so much about freedom, but about practical ability, or choice – assuming that we don't have a choice if we have only one option. Given our practical inability to affect the past or the laws, it seems that we have no practical abilities to do otherwise, and no choices to make, unless the world is indeterministic.

And would indeterminism help? Some interpretations of quantum physics suggest that the dynamics of our world is thoroughly random, subject only to statistical regularities. If our choices are random processes, having more than one option does not require a power to influence the past or the laws. But we don't think of our choices as random events. When we wonder what an agent should do, we don't wonder how some chance process that determines the agent's behaviour should go. The normative 'should' does not even seem appropriate if we're talking about the outcome of a chance process. We take for granted that the agent has a kind of control over their choice that is hard to reconcile with the idea that the choice is entirely random.

We are led to the abstruse conclusion that an agent's choices are neither determined nor random. Somehow, they impinge on the physical world from the outside, from a special realm that is not subject to natural laws.

6 The picture

Let's review the picture that has emerged from the four puzzles. In sections 2–4, we met some arguments suggesting that an agent must have perfect information about their options. Since people generally don't have perfect information about which ordinary acts they can perform, this led us to posit special acts of “willing” whose availability is never in doubt. The considerations in section 4 also suggested that willings are special in that there is no epistemic gap between deciding to act and acting: if an agent decides in favour of a willing, they can be rationally certain that they succeed.

I use 'willing' as a technical term, as a placeholder for whatever these special acts might be. Some authors prefer 'trying' or 'intending' or 'deciding'. The label is not important. We shouldn't assume that any word of ordinary English unambiguously captures the target.

Willings don't seem to be individuated physically or functionally. For any non-trivial physical or functional condition, one can easily imagine an agent who is rationally unsure whether they can bring about an event that satisfies the condition. Willings seem to inhabit a different kind of space, a luminous realm that is directly and infallibly accessible to the agent.

The other two puzzles, discussed in section 5, point in the same direction. They suggest that the acts an agent can choose should be understood as “interventions” into the physical world from a realm that is not subject to the natural laws.

This is the picture that has emerged. I'm not saying that it is correct. I'm saying that something like this appears to be implicit in our thinking about agency. Each of the puzzles began with assumptions that seem plausible, that seem to be part of our conception of agency. The picture is what we get when we follow these assumptions.

Some may find the picture appealing. To me, it looks crazy. Nothing in current science suggests that human agency involves a special realm of non-physical events. Even if there were such a realm, why would agents have perfect information about it? Where would that information come from? Is it

a priori? Revealed by a special, infallible kind of perception? By an infallible intuition? No answer looks remotely plausible. The interventionist aspect of the picture looks even wilder. Real choices are not uncaused interventions. They have a causal explanation. Jack takes the bus rather than the limousine because he is aware of his pecuniary situation and doesn't want to bankrupt himself. His choice is fully explained by his goals and information. For all I know, it may also be determined by the distant past and the laws: determinism is a live possibility in contemporary physics. At the very least, I can conceive of agents like Jocasta (from the previous section) who rationally *believe* that their world is deterministic. If we evaluate options as undetermined interventions, every possible choice entails the falsity of determinism. We would reach the bizarre conclusion that every rational agent should deny determinism, irrespective of their evidence.

I'm not sure *how* crazy the picture is: whether it is merely false or downright incoherent ("not even false", in Pauli's memorable words). It's tempting to think that the epistemic side of the picture is refuted by the conceivability of Frankfurt cases. Suppose an agent's deliberation process is monitored by a demon, determined to prevent a particular choice if they see it coming. This seems to make the choice practically unavailable. You can't even *will* to raise your arm if a demon would prevent you from willing. But you needn't know that the demon is there. If an agent has inconclusive information about whether they are in a Frankfurt case – and don't we all, if we put on our philosophy hats? – they have inconclusive information about their options, even if the options are willings.

The point could also be made with phobias. Following [Lehrer 1968], it is widely thought that agents with severe phobias can't even *try* or *intend* or *will* to perform certain acts that would otherwise be available: an arachnophobe can't will to touch a spider.¹³ Presumably, however, an agent may have inconclusive information about whether they suffer from arachnophobia. If so, they have inconclusive information about the available willings.

Here, however, the causal side of the picture might come to the rescue. If willings are instantaneous and uncaused interventions, perhaps no demon could "see them coming". If an agent represents their options as uncaused interventions, they might conclusively rule out being in a Frankfurt case. The same is true for phobias, if these are assumed to prevent the relevant choice shortly before or while it occurs. If willings are instantaneous and uncaused interventions, no psychological process could foresee and prevent them or stop them in their tracks.¹⁴

If the picture is incoherent, I suspect the incoherence lies not in its epistemic, but in its causal element, in the idea of rational acts that are neither determined nor random. Libertarians about free will have filled many volumes trying to explain how this might be possible. I am not convinced that they have succeeded. The task is especially challenging for the picture I have described because it

13 Personally, I'm not convinced by this model of phobias. Phobias are a type of fear. Like other fears, they arguably work by motivating the agent to avoid the relevant acts. They don't remove options, but associate the options with very low utility. An arachnophobe *can* will to touch a spider, in the practical sense of 'can'; they just don't want to. Compare [Caplan 2006] and [Alvarez 2013] for similar models of compulsive behaviour.

14 We have to be careful about timing. The conclusion we've reached from the first two puzzles is that an agent has perfect information about what willings they can realize right now. They need not have perfect information about what they can do in the future, if only because they may not be sure whether they'll still exist. The puzzle from section 4 may appear to suggest that a deliberating agent must have perfect information about their *future* options, about what they can do at the end of the deliberation. But this isn't so. It suffices that the agent is sure about what they can eventually choose conditional on their continued survival and agency.

posits a libertarian kind of control even for cases like Jack's, where all the reasons point in favour of one option and against the other. Jack *could* take the limo. He takes the bus not because it's the only option but because the alternative is a foolish waste of money. How can we treat him as a rational agent and yet assume that his reasons, up to the point of decision, somehow leave open what he will choose?¹⁵

7 Armchair robotics

Let's take a short break and think about how we might design a robot. (Bear with me. The point of the exercise will become clear soon.) The robot we are designing has a central processor and a database to store goals and information about the world. The robot also has a motor system that can be activated by the central processor. Imagine wires running from the robot's wheels and limbs to the central processor. Depending on the electrical current on the wires, the wheels and limbs move in different ways. We have to design an algorithm that determines which electrical signal is sent down which wires, based on the contents of the robot's database.

Here's one approach we could take. First, we figure out what movement each signal causes. Suppose activating the red wire normally causes the robot's left arm to rise. Whenever the database indicates that raising the left arm would bring the robot closer to its goals, we let the motor interface activate the red wire. Similarly for the other signals.

This would work well if activating the red wire always caused the left arm to rise. But it doesn't. At some point, for example, the robot might be positioned next to a wall, so that it can't raise its left arm. The signal in the red wire would now cause the arm to press against the wall. Doing so might be a good idea, or it might be a terrible idea, depending on what the robot wants to achieve and what it knows about the wall. We want to send the signal if *pressing against the wall* is expected to lead to good outcomes.

This kind of case is all too common. Depending on the incline of the floor and the wind, the same motor signal might cause the robot to accelerate, maintain pace, or slow down. The robot may not have perfect information about which of these conditions obtain, and therefore about what would happen if a given signal were sent.

Here's a better approach.¹⁶ We introduce primitive elements $X_1, X_2, X_3, \dots$ into the robot's database and decision algorithm. I'll call them *motor commands*. They serve as the options from which the robot chooses. We design the robot so that the red wire is activated iff the robot chooses X_1 , the blue wire iff it chooses X_2 , and so on. To the robot's database, we add beliefs about what is likely to happen as the result of the motor commands: that X_1 normally causes the robot to move forward, X_2 causes its left arm to go up, and so on. We can also add information about how these outcomes depend on the circumstances: if the robot's left arm is obstructed, X_2 causes the arm to exert force against the obstruction. And so on.

Motor commands are not descriptions of real acts. X_2 does not mean *I lift the left arm*. It doesn't

¹⁵ Libertarians often restrict their account to options between which the agent is torn, and sometimes locate the relevant "intervention" at a time before the agent's choice. This may help to make the account more plausible, but it isn't compatible with the picture that I think is implicit in our conception of agency.

¹⁶ I've advocated this approach in [Schwarz 2021].

mean *I press against the wall*. I doesn't mean *I send current down the red wire*. It is a primitive element in the robot's database and decision algorithm, logically independent (in the robot's doxastic space) of any real acts or events. Its purpose is not to represent an aspect of reality, but to allow for a simple and versatile interface between the robot's information and goals on the one hand and its motor system on the other.

In normal situations, the robot may be confident about what would happen if it chooses X2. If there is no reason to think that the left arm is obstructed, we (and the robot) might describe a choice of X2 as a choice to lift the left arm. Alternatively, we might describe X2 as sending a current down the red wire. If the robot happens to know about its inner workings, it might also think of X2 in this way. But our design does not require such knowledge, and it does not require a perfect association between motor commands and electrical signals: our design still works if a choice of X2 doesn't lead to activation of the red wire if the robot's battery is low or the temperature extreme.

This is – roughly – how I would design a robot. It is – roughly – how actual robots are designed. I suspect it is – roughly – how evolution has designed us.

Now imagine an agent who works like this. Call him J. J isn't sure whether his left arm is obstructed. If it isn't, lifting the arm would be a good idea, given his goals and information. If the arm is obstructed, it would be better to move back a few steps, but it wouldn't be a disaster if he tried to lift it. What should he do?

We may not want to say that he should lift his arm. J himself certainly shouldn't evaluate the relevant option as *lifting the arm*: this would disregard the possibility that the arm is obstructed. J evaluates his options as motor commands, deciding that X2 is his best option, better than X1 or X3, etc. But how should *we* think about his choice? We have no idea how he represents his options, and it wouldn't help us if we did: 'X2' means nothing to us.

What we can say is that J should choose *whichever motor variable he expects to lift his arm*, or something like that. We might introduce a shorthand: J should *try to lift his arm*, or *will to lift his arm*. On this understanding, 'trying' or 'willing' doesn't describe a special type of act. It hints at something only the agent can properly represent to themselves.

Even J, however, need not think of his options as motor commands. If X2 normally causes his arm to go up and is likely to do so now, he, too, may conceptualize X2 as trying (or willing) to lift his arm.

As J goes through his life, he may often be unsure about which overt acts he can perform. He doesn't know if his arm is obstructed, if he has enough energy, if the floor is slippery. He takes the different possibilities into account when he chooses a motor variable. Would he also need to consider if he can realize a particular motor command – if he can "will" to lift his arm? Perhaps not.

What would a situation look like in which J can't realize, say, X2? From an objective point of view, there are no such situations, as 'X2' doesn't designate a real kind of act (or event or proposition). There are situations in which J can't send the electrical signal associated with X2, because he is turned off. But that's not a possibility he needs to take into account when making a choice. Other situations in which he can't send the signal are Frankfurt cases where the relevant choice would be anticipated and prevented. One might argue that J should treat these as scenarios in which he can't realize X2. But Frankfurt cases are not only unusual and strange, it's also not clear what an agent

should do if they suspect they are in such a case. Suppose J would like to lift his arm, by choosing *X2*, but suspects a demon would prevent the choice. What should he do? One is tempted to say that he should *try to choose X2*, and that he should take the possibility of failure into account. But this would only introduce another layer of options, where the same problem would arise: What should he do if he suspects that a demon would prevent the decision to try to choose *X2*?

The upshot is that an agent like J has little use for a capacity to be unsure about the available motor commands. He may take it for granted that all his motor commands are on the table. He may be designed so that he is always certain about the “willings” he can exercise.

Now let’s think about how J might evaluate his options. What does he believe would happen if he chose *X2*? If his cognitive system works as I’ve suggested, his database will indicate what might happen as a result of choosing *X2*. It might tell him, as a general rule, that his arm will tend to go up unless it is obstructed. It might also tell him that his arm is currently not obstructed, so that he can infer that his arm would probably go up. But should his database tell him how *X2* would have come about? Perhaps not.

There are several reasons for letting his database fall silent on this question. Some have to do with computational economy and physical asymmetries. I want to focus on a simpler point. We don’t want J to “reason backward” when he evaluates his options. Suppose he believes that if he were to choose *X2* then the past would be different in a certain respect, and he lets this belief guide his choice. He might then opt for *X2* merely because he prefers the alternative past associated with *X2*. We wouldn’t want our robot to reason this way, given that the robot has no real influence on the past. For the same reason, we wouldn’t want our robot to think that different choices come with different laws of nature.

It might be best to design J so that he represents his motor commands as interventions, with (more or less) predictable consequences, but no predictable causes. He wouldn’t thereby represent the motor commands as random – as outcomes of a well-defined chance process. No, nothing at all would be said about their origin. They would seem to interfere into the world’s causal structure from the outside, from an apparent realm where causal questions don’t arise.

8 Unravelling the mystery

In the first half of this paper, I have argued that a curious picture seems to lie behind our conception of agency – a picture in which the immediate objects of choice are special acts (“willings”) that intervene into the world’s causal structure from the outside and whose availability is infallibly revealed to the agent. The picture looks indefensible. In the previous section, we saw how it may nonetheless be largely correct. If our cognitive system is designed the way I’ve described, then our perspective involves a luminous dimension of options that do intervene into the world’s causal structure from the outside.

A caveat. The architecture I have described is severely idealised. Real humans have control over a fine-grained and high-dimensional space of body movements and an equally rich space of mental acts, including the formation of commitments and plans. An efficient handling of this richness calls for a more sophisticated design. I won’t enter into these complications, except to note that they might explain some respects in which the picture I have described is at odds with our experience.

In conscious deliberation, we rarely think of our options as willings or tryings, or even as body movements. In a restaurant, I might deliberate between *pizza* and *pasta*. Favouring the pasta, I might choose a string of words to express my decision, but this is typically sub-personal, as is the choice of muscle movements in my tongue and lips that produce the words. My decision to have pasta makes me confident that I will indeed be having pasta, but I recognize the possibility of failure: the restaurant could be out of pasta, the waiter could refuse to serve us, my muscles could fail to obey me. I could have accounted for these possibilities in my deliberation: I could have asked whether I should *try to have pizza*. But the costs of the failure possibility are essentially the same for pizza and for pasta, so nothing is lost by ignoring them. I could also have asked whether I should *try to have pizza by uttering such-and-such words* or even *by moving my mouth in such-and-such manner*. But it's easy to see why our cognitive system would delegate these choices to "lower-level" systems: the possible costs and benefits of the realistic options are essentially on a par, and don't affect whether I should have pizza or pasta. (It might be otherwise if the pasta had a long and difficult-to-pronounce name and I cared about not embarrassing myself.)

Let's return to the simple, non-hierarchical picture. The picture matches J's perspective. But is it *true* of J? Does J have the practical ability to perform special acts that intervene in the world's causal structure and whose availability is infallibly known? Of course not. The picture is not true of J, it is not true of us, and perhaps not of any possible agent. Can we tidy it up? Can we fix our picture of agency so as to make it correct?

The task isn't easy. The willings in the centre of the picture are fictions. When J "chooses X2", or "wills to lift his arm", what happens in reality is that his decision module sends an electrical signal down the red wire and updates his database so that X2 becomes certain. Neither we nor J could sensibly replace the fictitious willings in our reasoning about his choices with any such real events. We can't practically do it, for lack of physiological knowledge. But we also can't do it unless we want to drastically revise our conception of agency. Electrochemical events in an agent's decision system are neither uncaused interventions nor is their availability directly accessible to the agent. We would need a radically different model.

I don't know if there is an alternative to the picture I have described that is both workable and correct. I suspect the only correct alternative would represent agents as physical systems. We would gain in empirical accuracy, at a great loss in tractability. (And how would this work for our own deliberation?) Perhaps the best we can do is to play along with the fiction. We'll run into puzzles. We may wonder how an agent could have infallible information about their options, or how to make sense of rational but uncaused interventions. We may be stumped by cases like Jocasta's. But if we understand that the picture is just a picture, perhaps we needn't be too worried about the puzzles it raises.

9 Postscript: The epistemology of ability

I should say a little more on the topic of this volume: the epistemology of ability. In the picture I've sketched, agents have perfect knowledge about the available motor commands. No empirical or non-empirical basis is needed for this knowledge. What about knowledge of more ordinary abilities?

How do we know that we can, say, raise our left arm, assuming we are designed like J?

The question has many readings, if only due to the polysemy of ‘can’. Let’s focus, as a beginning, on the “specific”, “transparent”, and “pure” reading, where we’re concerned with what we can do intentionally, in the present circumstances, even if it might go against our inclinations or relevant norms.¹⁷ In essence: how could J know that choosing *X2* would raise his arm? More colloquially, how could he know that he would succeed if he tried (or decided, or willed) to raise his arm?

There are also different conceptions of knowledge. Suppose a skeptic challenges J’s claim to knowledge, by pointing out that his (J’s) left arm might have been amputated last night, and that he might now be hallucinating the arm. If J doesn’t know that he has a left arm, he plausibly can’t know that *X2* would raise the arm. In response, one might claim that J *sees* that he has a left arm, and that seeing is a kind of knowing. Fair enough. But the skeptic still has a point. Suppose before J opened his eyes today, he had reason to suspect that his arm has been amputated and that he would have a realistic hallucination of it. Suppose he rationally gave probability 0.8 to this hypothesis. How should this assessment change when he opened his eyes and (as we might put it) saw his arm? Answer: it should remain unchanged at 0.8. In general, J’s visual experience does not affect his rational credence in the amputation/hallucination hypothesis. If his prior credence in the hypothesis was 0.00001 (as it might well have been if he had no special reason to take the hypothesis seriously), his posterior credence should still be 0.00001. In that sense, the skeptic is right: J cannot rule out the amputation/hallucination hypothesis, even if he can see that he has arms. What we should say, against the skeptic, is that not all open possibilities are on a par: absent special reasons, J may rationally give negligible credence to the amputation/hallucination hypothesis.

Granting that he has arms, how could J be rationally confident that his left arm would go up if he tried to raise it, if he chose *X2*? There are no strict *a priori* connections, in J’s database, between motor commands and ordinary events. There are countless error possibilities. The arm would not go up if it is tied down or too close to a wall. Even if all such prerequisites are in place, there’s no guarantee (according to J’s database) that the arm will go up.

Could affordance perception plug the gap? Some hold that we perceive the objects in our environment as affording certain actions: we see a door knob as *grabbable* (*by us*). Assuming that something is grabbable by an agent only if the agent can grab it, it would follow that we can see that we can grab the door knob. One might also posit a special internal sense of our body and the actions it affords. Either kind of perception might tell J that he can raise his arm. (Maier, this volume, is sympathetic to these ideas.)

As before, I don’t think this gets to the heart of the challenge. Suppose you have just undergone surgery, and you are rationally unsure whether you can raise your arms. Your internal sense of what your body affords can’t tell the difference, nor can your eyes: the door knob would look just as grabbable either way. Affordance perception should not affect your credence in whether the surgery was a success. In that sense, it provides no information about your ability. The same is true for J. If he woke up with credence 0.0001 in the error scenarios, any affordance perception he might have should leave his credence unchanged.

17 See [Maier 2018] for the distinction between “specific” and “general”, [Schwarz 2020] for the distinction between “transparent” and “effective”, and [Kratzer 1977] for the distinction between “pure” and “impure”, though not with these labels.

I don't deny the reality of affordance perception. It may well be that our visual system somehow directly "tells us" that an object is grabbable, more or less as it "tells us" that we have arms. But I deny that we can rationally take the testimony of our senses as a direct foundation for our beliefs. In Bayesian terms, I deny that we may rationally conditionalize on what our senses tell us. The psychological story may be simple and direct, the epistemic story is not. Here I agree with Schoonen and Bruineburg, this volume, although I suspect that my version of the epistemic story is different from theirs.

The epistemic story I favour is the standard Bayesian story of induction. When I introduced motor commands, I suggested that our robot's database should specify what might happen as a result of the various commands: that *X2* tends to raise the left arm, except if a wall is in the way, in which case it would cause the arm to press against the wall, etc. We might have explicitly programmed all this information into the robot's database. But that would be cumbersome, inflexible, and easy to get wrong. Much better to let the agent discover the associations between motor commands and ordinary events.

The most basic method of discovery is trial and error. We might build an agent so that it randomly chooses among the available motor commands in the early days of its life, observing their effects. In this way, *J* could have observed that *X2* tends to move his left arm, and not his right arm or the nearby table, unless these are connected to his left arm. He could have observed that *X2* does not move his arm through walls and other barriers, and so on.

There is no logical guarantee that past associations will hold in the future. Accordingly, *J* should always reserve some credence for the possibility that *X2* won't lead to the left arm going up next time, even if there's no obstacle in the way. But he needn't worry too much about this possibility. Rational credence can (and should) be biased towards worlds in which past regularities persist into the future.

When *J* has observed his arm going up after issuing *X2*, he initially learns about a "specific" (and "effective") ability: that he was able to raise his arm on this occasion. The inductive generalisation gives him knowledge of a "general" ability: that he is able to raise his arm under such-and-such conditions. Knowing that the right conditions obtain, he can infer that he has the "specific" (and "transparent") ability even if the ability isn't exercised.

There are many other ways to learn about one's abilities. We can, for example, generalize from observed abilities to different abilities (see Kikkert, this volume). If *J* has never lifted an 18 kg weight, but he has lifted a range of lighter and heavier weights, he may be rationally confident that he can lift the 18 kg weight, given his observation that his abilities generally don't have weird gaps. Witnessing the success and failure of others may also help. If *J* sees that similar agents easily lift 20 kg weights, he may infer that he can do the same – that he, too, would lift the weight if he chose the right motor command.

More specialized sources of evidence may also play a role. Massin (this volume) points out that our experience of the effort required to perform an act helps us gauge our general capacities. If lifting 20 kg feels easy, you have reason to think that you could lift a 25 kg weight. Similarly, Spener (this volume) suggests that "metacognitive" feelings of confidence help gauge one's capacities. If you feel confident that you are solving a calculus problem correctly, you have reason to think that you can solve even harder problems. I am happy to grant these points. But I would insist that the relevant

feelings do not provide a self-standing epistemic basis for the relevant knowledge. It is a contingent empirical fact that if lifting 20 kg feels easy then lifting 25 kg (or even 20.1 kg) is possible.

The story I have sketched is empiricist and *actualist*, in the sense of Maier (this volume): Our knowledge of what we can do is based not on direct insight into unexercised abilities, but on mundane evidence of occurrent events – most importantly, of past success and failure.

This part of my story resembles the one in [Vetter 2024]. Vetter also suggests that our knowledge of our abilities is largely based on experience of past exercise. According to Vetter, however, the relevant experience consists in an observation of the act combined with an experience of agency. It is not clear to me how this could reveal to our cognitive system what it must do to perform the act – knowledge that seems required for transparent (“robust”) ability. It’s also not clear how it could ground knowledge of which acts we are able to perform on a given occasion: how do we know that it is ever in our power to do something that we don’t actually do?

Here my story involves non-empiricist and possibilist elements. The agents I have described always know what willing or motor command they have issued. They know this not on the basis of sensory feedback, but merely by having issued the command. They also know, without any relevant evidence, that alternative commands are available. In the architecture I have described, knowledge of the available motor commands comes for free.

This part of my story comes closer to that of Steward, in [Steward 2012] and the present volume. Like Steward, I think we are implicitly committed to a libertarian picture of agency. The commitment is not an ordinary belief, justified by pertinent evidence, but a crucial part of how we structure our thinking. Unlike Steward, I have no hope that it is true.

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